



UNIVERSIDAD NACIONAL AUTONOMA DE MEXICO

FACULTAD INGENIERIA

BASES DE DATOS

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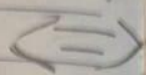
Tarea 10

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A	X	B	Y
7	2	6	11
3	4	9	15
10	7	2	4
1	12	2	11

B	W	D	Y	A	Z
2	5	6	11	1	30
4	7	8	4	7	8
9	10	11	28	5	12



1) $R_1 \times R_2$

2) $R_2 \bowtie R_1$

3) $R_1 \bowtie ((R_1.A > R_2.Z \text{ or } R_1.A >= R_2.W) \text{ and } R_1.Y = R_2.Y) R_2$

1) $R_1 \times R_2$:

$R_1.A$	$R_1.X$	$R_1.B$	$R_1.Y$	$R_2.B$	$R_2.W$	$R_2.D$	$R_2.Y$	$R_2.A$	$R_2.Z$
7	2	6	11	2	5	6	11	1	30
7	2	6	11	4	7	8	4	7	8
7	2	6	11	9	10	11	28	5	12
3	4	9	15	2	5	6	11	1	30
3	4	9	15	4	7	8	4	7	8
3	4	9	15	9	10	11	28	5	12
10	7	2	4	2	5	6	11	1	30
10	7	2	4	4	7	8	4	7	8
10	7	2	4	9	10	11	28	5	12
1	12	2	11	2	5	6	11	1	30
1	12	2	11	4	7	8	4	7	8
1	12	2	11	9	10	11	28	5	12

2) $R_2 \bowtie R_1$:

A	X	B	Y	W	D	Z
1	12	2	11	5	6	30

Select

FROM R_2

NATURAL JOIN R_1

3) $R_1.A > R_2.Z \text{ OR } R_1.A >= R_2.W \text{ AND } R_1.Y = R_2.Y$

$R_1.A$	$R_1.X$	$R_1.B$	$R_1.Y$	$R_2.B$	$R_2.W$	$R_2.D$	$R_2.Y$	$R_2.A$	$R_2.Z$
7	2	6	11	2	5	6	11	1	30
10	7	2	4	4	7	8	4	7	8

CREATE TABLE R_1 (A Int, X Int, B Int, Y Int);

CREATE TABLE R_2 (B Int, W Int, D Int, Y Int, A Int, Z Int);

Insert Into R_1 (A, X, B, Y) values

(7, 2, 6, 11), (3, 4, 9, 15), (10, 7, 2, 4), (1, 12, 2, 11);

Insert Into R_2 (B, W, D, Y, A, Z) values

(2, 5, 6, 11, 1, 30), (4, 7, 8, 4, 7, 8), (9, 10, 11, 28, 5, 12)

Select

FROM R_1

CROSS JOIN R_2

Select

FROM R_1

JOIN R_2

ON ($R_1.A > R_2.Z \text{ OR } R_1.A >= R_2.W$) AND ($R_1.Y = R_2.Y$)